UTS

Uber Data Analytics – Report (B)

Python Programming for Data Processing

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Assessment 2

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Introduction

This report provides a comprehensive overview of the data preparation and initial analysis of a dataset from Uber, based on their rides. The main objective is to clean and structure the data, identifying key data quality issues, and then generate a meaningful summary including statistics and a correlation analysis.

For this instance, this analysis could reveal interesting correlation between variables to answer questions like if a longer trip means a higher customer satisfaction, if a certain payment method is more likely to be used in shorter trips or in longer ones, etc.

Once these insights are identified, we could optimize Uber's decision-making process, based on cost, customer rating, or other.

Dataset

The dataset used for this analysis is a publicly available file from Kaggle (open-source platform), named ncr_ride_bookings.csv. It contains 150,000 entries for 21 attributes. The data provides specific information for each ride, including identifying information like booking ID and customer ID, as well as trip-related variables such as booking date, time and status. It also includes key performance indicators (KPIs), including booking value, ride distance and satisfaction metrics like driver ratings and customer ratings. The dataset's structure is illustrated below.

```python
# Print the general information about the dataframe, to evaluate if it is suitable for the assignment
print("Dataframe")
df.info()
```

```
Dataframe
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150000 entries, 0 to 149999
Data columns (total 21 columns):
 #   Column                              Non-Null Count   Dtype
---  ------                              --------------   -----
 0   Date                                150000 non-null  object
 1   Time                                150000 non-null  object
 2   Booking ID                          150000 non-null  object
 3   Booking Status                      150000 non-null  object
 4   Customer ID                         150000 non-null  object
 5   Vehicle Type                        150000 non-null  object
 6   Pickup Location                     150000 non-null  object
 7   Drop Location                       150000 non-null  object
 8   Avg VTAT                            139500 non-null  float64
 9   Avg CTAT                            102000 non-null  float64
 10  Cancelled Rides by Customer         10500 non-null   float64
 11  Reason for cancelling by Customer   10500 non-null   object
 12  Cancelled Rides by Driver           27000 non-null   float64
 13  Driver Cancellation Reason          27000 non-null   object
 14  Incomplete Rides                    9000 non-null    float64
 15  Incomplete Rides Reason             9000 non-null    object
 16  Booking Value                       102000 non-null  float64
 17  Ride Distance                       102000 non-null  float64
 18  Driver Ratings                      93000 non-null   float64
 19  Customer Rating                     93000 non-null   float64
 20  Payment Method                      102000 non-null  object
dtypes: float64(9), object(12)
memory usage: 24.0+ MB
```

Columns for identification and temporality

Date, time, booking ID and customer ID are the foundation of the analysis. They allow us to track the consumer behaviour over time, identify seasonal trends and make sure that we're working with unique data of each trip.

Column for key metrics

Booking value, ride distance, driver ratings and customer ratings are essential for business analysis. These columns help us answer questions regarding performance and customer satisfaction, like the average trip value, how far customers ride in average, how is the service perceived by customer, do they value it or not, etc.

Columns for segmentation

Vehicle type and payment method help us segmentate the data to understand user's preferences. For example, we could analyse if the clients that use eBike have a different valuation than the ones who use GoSedan, or if a payment method is preferred.

```python
# Display the first 5 rows of the dataset
print("First 5 Rows")
print(df.head().to_markdown(numalign="left", stralign="left"))
```

First 5 Rows

	Date	Time	Booking ID	Booking Status	Customer ID	Vehicle Type	Pickup Location	Drop Location
0	2024-03-23	12:29:38	"CNR5884300"	No Driver Found	"CID1982111"	eBike	Palam Vihar	Jhilmil
1	2024-11-29	18:01:39	"CNR1326809"	Incomplete	"CID4604802"	Go Sedan	Shastri Nagar	Gurgaon Sector 56
2	2024-08-23	08:56:10	"CNR8494506"	Completed	"CID9202816"	Auto	Khandsa	Malviya Nagar
3	2024-10-21	17:17:25	"CNR8906825"	Completed	"CID2610914"	Premier Sedan	Central Secretariat	Inderlok
4	2024-09-16	22:08:00	"CNR1950162"	Completed	"CID9933542"	Bike	Ghitorni Village	Khan Market

Avg VTAT	Avg CTAT	Cancelled Rides by Customer	Reason for cancelling by Customer	Cancelled Rides by Driver	Driver Cancellation Reason
nan	nan	nan	nan	nan	nan
4.9	14	nan	nan	nan	nan
13.4	25.8	nan	nan	nan	nan
13.1	28.5	nan	nan	nan	nan
5.3	19.6	nan	nan	nan	nan

Incomplete Rides	Incomplete Rides Reason	Booking Value	Ride Distance	Driver Ratings	Customer Rating	Payment Method
nan	nan	nan	nan	nan	nan	nan
1	Vehicle Breakdown	237	5.73	nan	nan	UPI
nan	nan	627	13.58	4.9	4.9	Debit Card
nan	nan	416	34.02	4.6	5	UPI
nan	nan	737	48.21	4.1	4.3	UPI

```python
# Display the last 5 rows of the dataset
print("Last 5 Rows")
print(df.tail().to_markdown(numalign="left", stralign="left"))
```

Last 5 Rows

	Date	Time	Booking ID	Booking Status	Customer ID	Vehicle Type	Pickup Location	Drop Location	Avg VTAT
149995	2024-11-11	19:34:01	"CNR6500631"	Completed	"CID4337371"	Go Mini	MG Road	Ghitorni	10.2
149996	2024-11-24	15:55:09	"CNR2468611"	Completed	"CID2325623"	Go Mini	Golf Course Road	Akshardham	5.1
149997	2024-09-18	10:55:15	"CNR6358306"	Completed	"CID9925486"	Go Sedan	Satguru Ram Singh Marg	Jor Bagh	2.7
149998	2024-10-05	07:53:34	"CNR3030099"	Completed	"CID9415487"	Auto	Ghaziabad	Saidulajab	6.9
149999	2024-03-10	15:38:03	"CNR3447390"	Completed	"CID4108667"	Premier Sedan	Ashok Park Main	Gurgaon Sector 29	3.5

Avg CTAT	Cancelled Rides by Customer	Reason for cancelling by Customer	Cancelled Rides by Driver	Driver Cancellation Reason
44.4	nan	nan	nan	nan
30.8	nan	nan	nan	nan
23.4	nan	nan	nan	nan
39.6	nan	nan	nan	nan
33.7	nan	nan	nan	nan

Incomplete Rides	Incomplete Rides Reason	Booking Value	Ride Distance	Driver Ratings	Customer Rating	Payment Method
nan	nan	475	40.08	3.7	4.1	Uber Wallet
nan	nan	1093	21.31	4.8	5	UPI
nan	nan	852	15.93	3.9	4.4	Cash
nan	nan	333	45.54	4.1	3.7	UPI
nan	nan	806	21.19	4.6	4.9	Credit Card

Data processing outcomes
For this task, I used Pandas (including NumPy) to analyse the data. This included column and row filtering, as well as generation of descriptive statistics and correlation analysis.

We will delete the columns that have extremely high percentages of null values, because they're not useful for general analysis. Even though they could be very important for specific analysis, they're not essential to analyse main key metrics for a business.

We also need to delete rows that contain information that is not relevant for analysis. For example, if a row's booking status is "no driver found", then the trip never happened. These rows don't contain key information as booking value or ride distance, or ratings. If we don't delete these rows, it would distort our results.
We delete: avg VTAT, avg CTAT, cancelled rides by customer, reason for cancelling by customer, cancelled rides by driver, driver cancellation reason, incomplete rides and incomplete rides reason.

Even after filtering no driver found trips, some rows have null values in key columns as booking value and ride distance. We can't calculate the average value of a trip or distance if these data don't exist. Deleting these rows ensure a more accurate analysis, with useful and complete information.

Statistical summary
We define key numerical columns: booking value, ride distance and ratings (customer and driver).

- Booking value: the average booking value is approximately $508.30 and a median of $414. It also has a high standard deviation of 395.81. The fact that the mean is higher than the median, suggests that the data may be slightly skewed by a small number of high-value trips (possible outliers).
- Ride distance: the average ride distance is 24.64 km and a median of 23.72 km. This indicated that most trips fall within a similar distance range, but with some longer rides influencing the mean. This is crucial information for understanding the typical behaviour of customers and drivers within Uber.
- Customer ratings: the average customer rating score is 4.4 with a median of 4.5 and standard deviation of 0.44.
- Driver ratings: the average driver rating score is 4.23 with a median of 4.3, which are slightly lower than customer ratings. This variable also has and a standard deviation of 0.44.

```python
# Generate summary statistics (for numerical columns)
print("--- Summary Statistics for Numerical Columns ---")
print(df_cleaned.describe().to_markdown(numalign="left", stralign="left"))
```

```
--- Summary Statistics for Numerical Columns ---
|       | Booking Value   | Ride Distance   | Driver Ratings   | Customer Rating   |
|:------|:----------------|:----------------|:-----------------|:------------------|
| count | 102000          | 102000          | 93000            | 93000             |
| mean  | 508.296         | 24.637          | 4.23099          | 4.40458           |
| std   | 395.806         | 14.0021         | 0.436871         | 0.437819          |
| min   | 50              | 1               | 3                | 3                 |
| 25%   | 234             | 12.46           | 4.1              | 4.2               |
| 50%   | 414             | 23.72           | 4.3              | 4.5               |
| 75%   | 689             | 36.82           | 4.6              | 4.8               |
| max   | 4277            | 50              | 5                | 5                 |
```

Correlation table

The correlation matrix for the main numerical attributes reveals a significant finding, a near-zero linear relationship between key variables such as booking value, ride distance and both rating variables. The correlation coefficient between booking value and ride distance is only 0.00517, which is a very low and unexpected result. In a typical ride application, one would expect a strong positive correlation between the trip distance and its cost. But the lack of this relationship suggests that other factors, potentially dynamic pricing models, vehicle type (luxury or standard), location-based fares or highway usage additional fee, have a more dominant influence on the final booking value, more than the ride distance itself.
Similarly, the correlation between ride distance and customer rating is only 0-00451, indicating that the length of a trip does not have a direct linear impact on customer satisfaction. One would expect that these variables have a positive correlation, as in longer private trips (instead of public transport use), the passenger tends to value even more the private ride service, however, the coefficient is extremely low to establish a relationship of correlation or causality.

```python
# Correlation table for the numerical columns
print("Correlation matrix for numerical columns")

numerical_cols = df_cleaned.select_dtypes(include=['float64', 'int64']).columns
correlation_matrix = df_cleaned[numerical_cols].corr()
print(correlation_matrix.to_markdown(numalign="left", stralign="left"))
```

```
Correlation matrix for numerical columns
|                  | Booking Value   | Ride Distance   | Driver Ratings   | Customer Rating   |
|:-----------------|:----------------|:----------------|:-----------------|:------------------|
| Booking Value    | 1               | 0.00517398      | -0.000248511     | -0.000286631      |
| Ride Distance    | 0.00517398      | 1               | -0.00187453      | 0.00451448        |
| Driver Ratings   | -0.000248511    | -0.00187453     | 1                | -0.00101035       |
| Customer Rating  | -0.000286631    | 0.00451448      | -0.00101035      | 1                 |
```

Data quality

The dataset, despite its size, presents several data quality challenges that must be adjusted for a more accurate analysis and use of data.

The most significant challenge is that even after filtering out “no driver found” trips, the driver ratings and customer rating columns still contains 9,000 missing values each. These missing values are a key obstacle, as they prevent a direct and comprehensive analysis of customer and driver satisfaction. It also represents a large portion of the data that needs to be handled.

On the other hand, the date and time columns are incorrectly stored as object (text, string) types, which complicates any time-series analysis to identify trends such as peak booking hours or seasonal demand fluctuations (i.e: public holidays or sports game day). It would be better to convert it to a date format.

Third, textual columns like booking ID and customer ID contain inconsistent formatting, especially because of the unnecessary double quotes (“), which would need to be cleaned before these columns could be used for advanced data analysis or data merging. This would make the data more proper and facilitate the data manipulation process.

```
# Identify Data Quality Issues
print("Null values ")
print(df_cleaned.isnull().sum().to_markdown(numalign="left", stralign="left"))

print("Data types")
print(df_cleaned.info())
```

```
Null values
|                 | 0    |
|:----------------|:-----|
| Date            | 0    |
| Time            | 0    |
| Booking ID      | 0    |
| Booking Status  | 0    |
| Customer ID     | 0    |
| Vehicle Type    | 0    |
| Pickup Location | 0    |
| Drop Location   | 0    |
| Booking Value   | 0    |
| Ride Distance   | 0    |
| Driver Ratings  | 9000 |
| Customer Rating | 9000 |
| Payment Method  | 0    |
Data types
<class 'pandas.core.frame.DataFrame'>
Index: 102000 entries, 1 to 149999
Data columns (total 13 columns):
 #   Column           Non-Null Count   Dtype
---  ------           --------------   -----
 0   Date             102000 non-null  object
 1   Time             102000 non-null  object
 2   Booking ID       102000 non-null  object
 3   Booking Status   102000 non-null  object
 4   Customer ID      102000 non-null  object
 5   Vehicle Type     102000 non-null  object
 6   Pickup Location  102000 non-null  object
 7   Drop Location    102000 non-null  object
 8   Booking Value    102000 non-null  float64
 9   Ride Distance    102000 non-null  float64
 10  Driver Ratings   93000 non-null   float64
 11  Customer Rating  93000 non-null   float64
 12  Payment Method   102000 non-null  object
dtypes: float64(4), object(9)
memory usage: 10.9+ MB
None
```

```python
# Check if there are any relevant outliers in numerical columns
print("Outliers")
print(df_cleaned[['Booking Value', 'Ride Distance', 'Driver Ratings', 'Customer Rating']].describe().to_markdown(numalign="left", stralign="left"))
```

```
Outliers
|       | Booking Value   | Ride Distance   | Driver Ratings   | Customer Rating   |
|:------|:----------------|:----------------|:-----------------|:------------------|
| count | 102000          | 102000          | 93000            | 93000             |
| mean  | 508.296         | 24.637          | 4.23099          | 4.40458           |
| std   | 395.806         | 14.0021         | 0.436871         | 0.437819          |
| min   | 50              | 1               | 3                | 3                 |
| 25%   | 234             | 12.46           | 4.1              | 4.2               |
| 50%   | 414             | 23.72           | 4.3              | 4.5               |
| 75%   | 689             | 36.82           | 4.6              | 4.8               |
| max   | 4277            | 50              | 5                | 5                 |
```

Conclusion

The data processing and exploratory analysis of the ncr_ride_bookings.csv dataset has been an important first step in preparing and cleaning the data for further analytics. The process included a few transformations from a raw dataset into a structured and more usable format by filtering out irrelevant data and identifying key variables for deeper analysis. A overall review of the descriptive statistics provided a clear overview of the central tendencies and variability of the dataset. For instance, the analysis of booking value and ride distance revealed that while a wide range of values exists, the mean and median are aligned, suggesting that most of the trips are consistent in both cost and length. The comparison of driver and customer ratings also provided an early insight into service quality perception, indicating that customers, on average, rate their overall experience slightly higher than the drivers are rated.

One of the most significant findings was the lack of a strong linear correlation between the main numerical variables.

Despite the initial preparation, the analysis also highlighted several critical data quality issues that must be addressed in subsequent stages. The presence of s substantial number of missing values is the most challenging step. Furthermore, the incorrect data types for date and time fields will need to be re-formatted for deeper time-series analysis.